


KTH ROYAL INSTITUTE
OF TECHNOLOGY

# PCC1: Lab 1

## Program a Medical Device

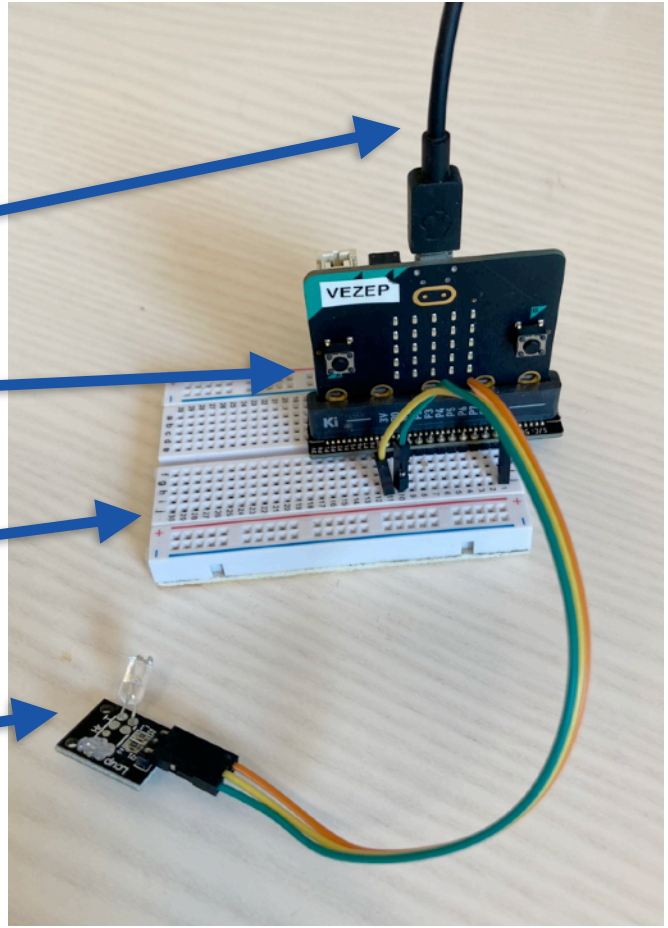
# Hardware

USB Cable (for Power)

Processing Unit

Breadboard

photoplethysmograph
(or your Lab 0 solution!)



# Connect to PPG circuit

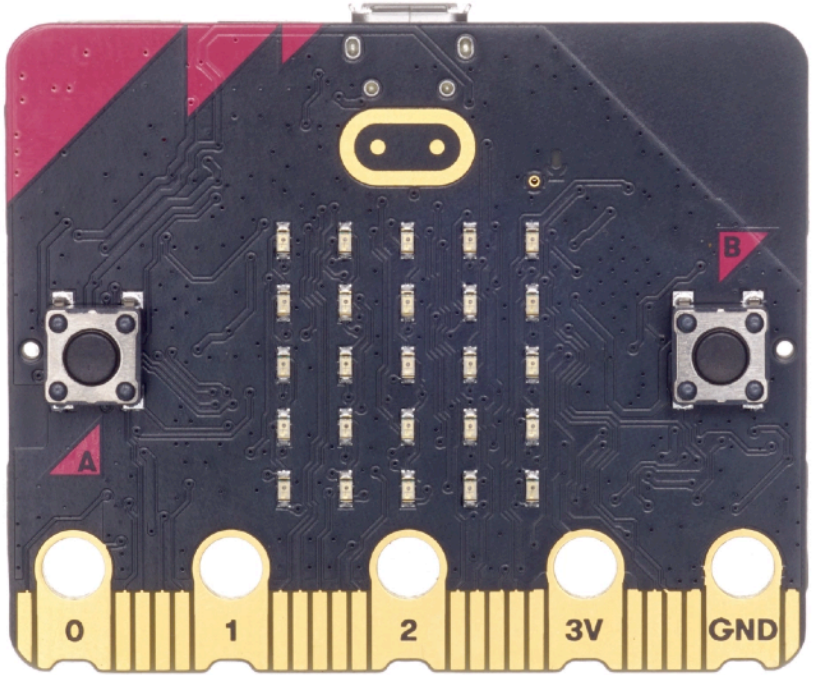
Connect your PPG circuit to:

3V

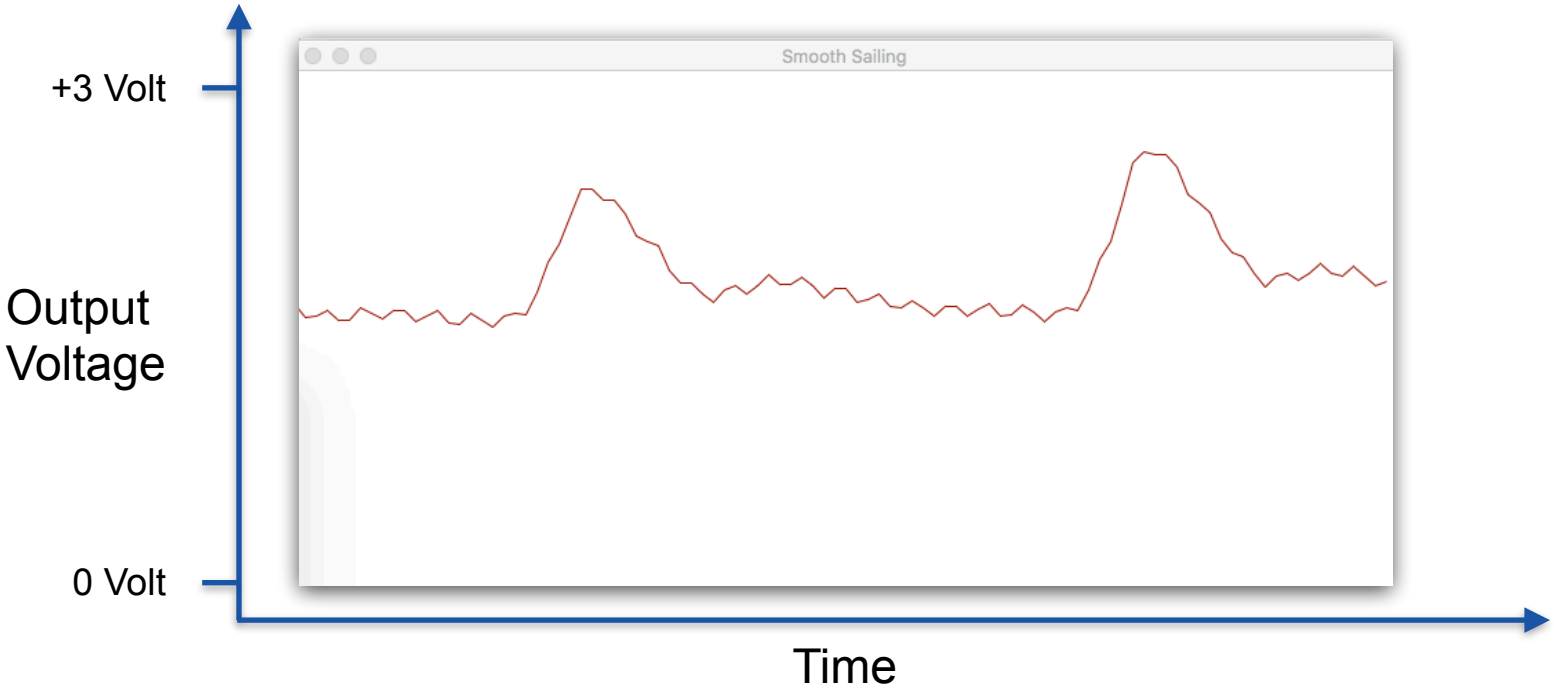
GND

PIN0

*PIN0 is an ADC*



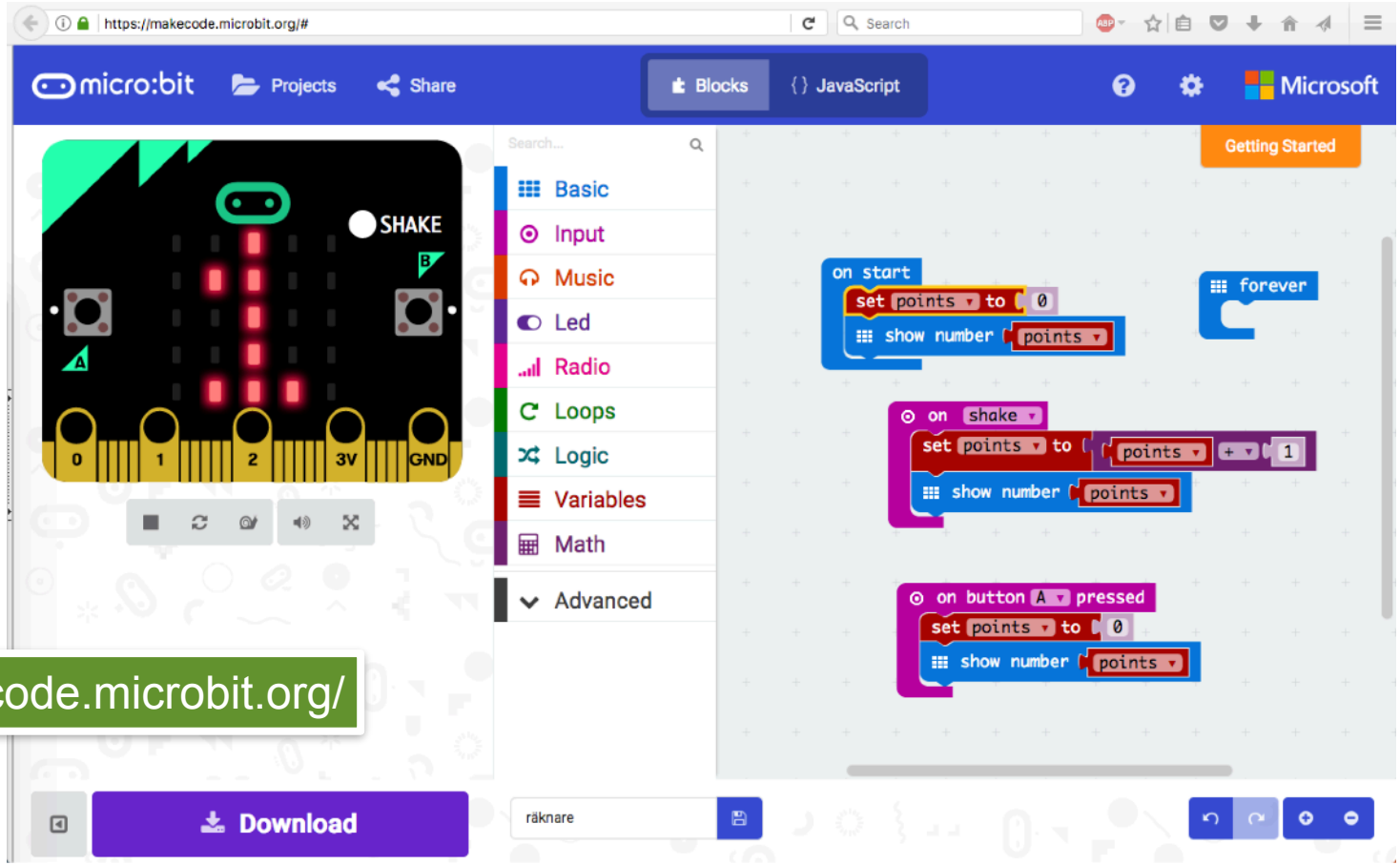
# Graph Drawing



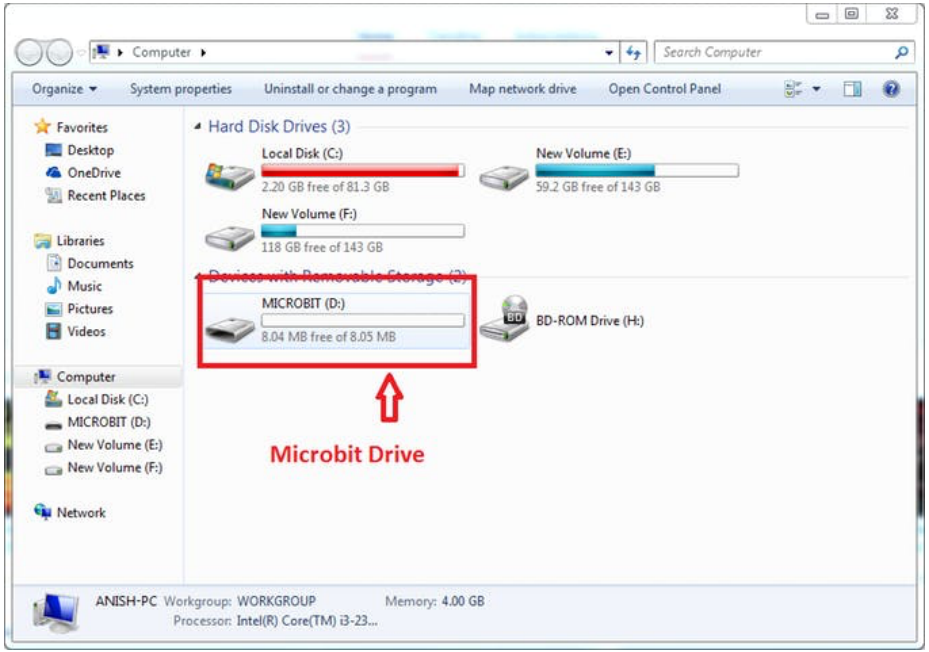
Step 1:
Test the Development Environment

Try the simulator

http://makecode.microbit.org/



# Step 2: Program the Micro:bit




Check that the Micro:bit drive appears

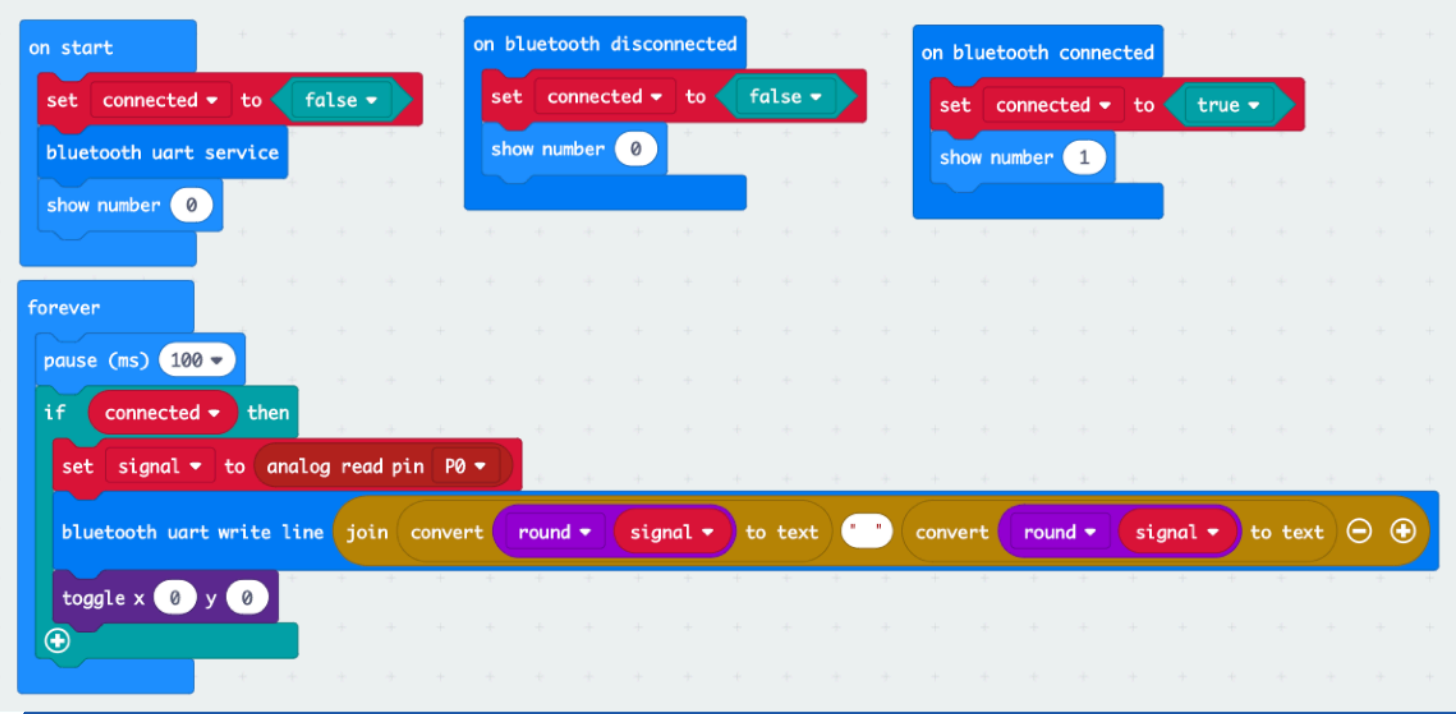
# Step 2: Program the Micro:bit (cont:d)

1. Connect it to the computer with the USB cable.
2. Wait.
3. Maybe you are asked to install updates. Do that.
4. Change the setting in Chrome:
    1. Click  in the top right corner
    2. Choose Settings / Advanced / Download
    3. Choose Ask where to save.
5. Go back to Development Environment
6. Click Download and save to the Micro:bit drive.



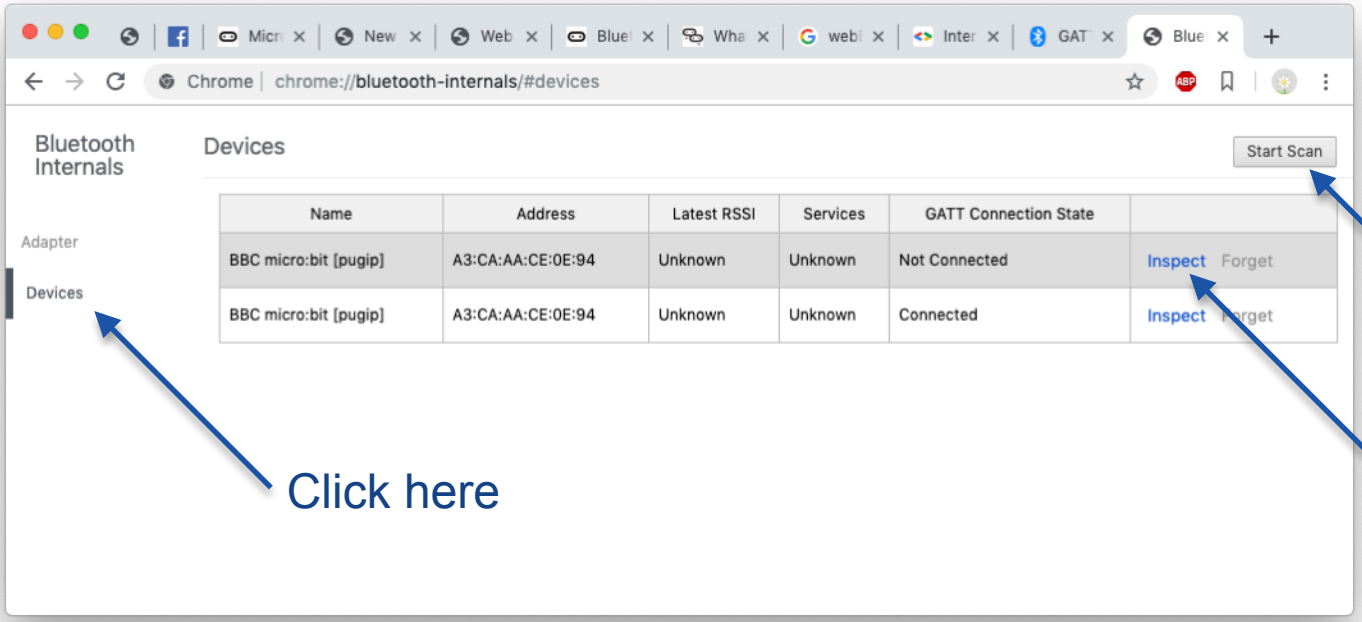

7. Test the code on the Micro:bit by shaking it and press button A.

# Step 3: A basic BLE program



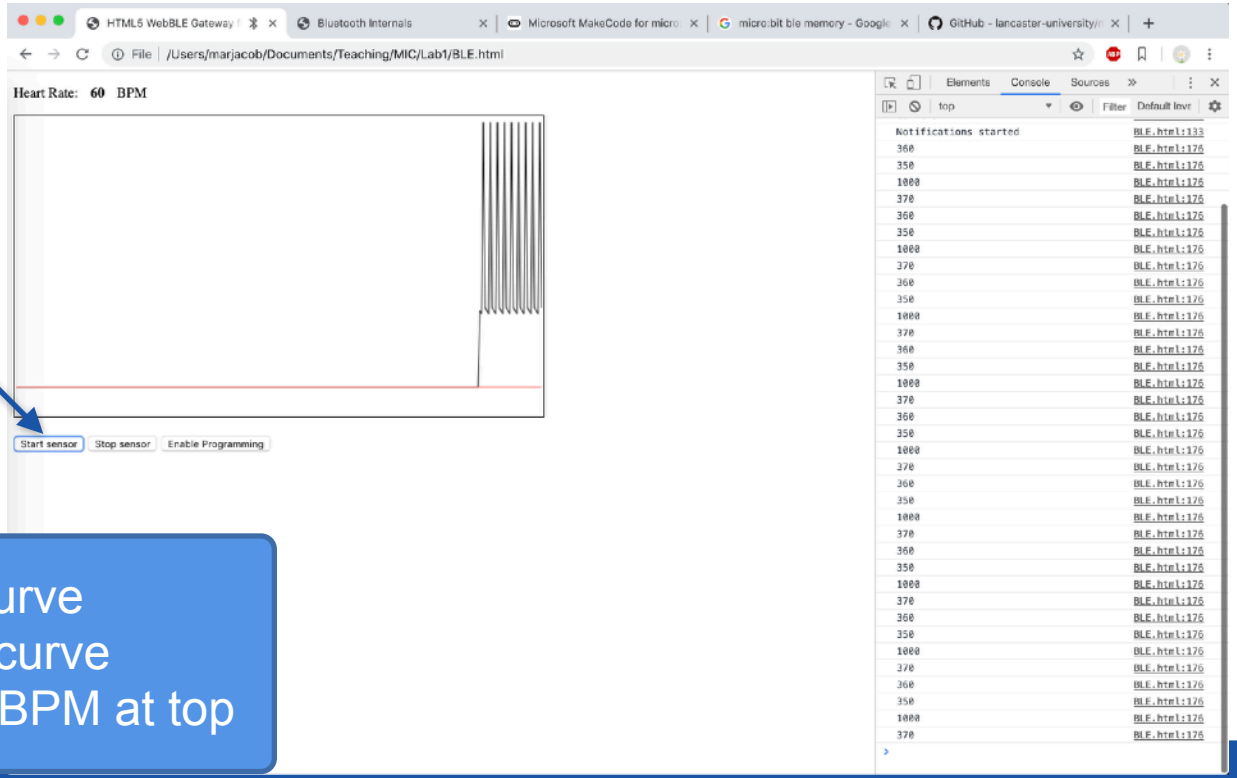
# Step 4: WebBLE

chrome://bluetooth-internals



# Step 4: Connect to the Web page in Chrome

**Click here to connect to the device**




First value is black curve
Second value is red curve
Third value show as BPM at top

# Step 5: Make the Code

Use the PPG and the Micro:bit

Make a device that can measure the heart beat (BPM)

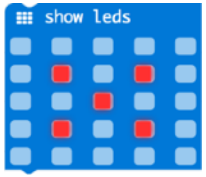
- Blink a LED (on the Micro:Bit) with the heart beat
- Connect it to a computer with Bluetooth and
- Calculate the BPM and display it with the WebBLE web script in Google Chrome
- together with the blood vessel graph.

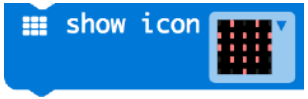
# Some Small Hints

Find the ‘time’:




Some commands are slow, such as:



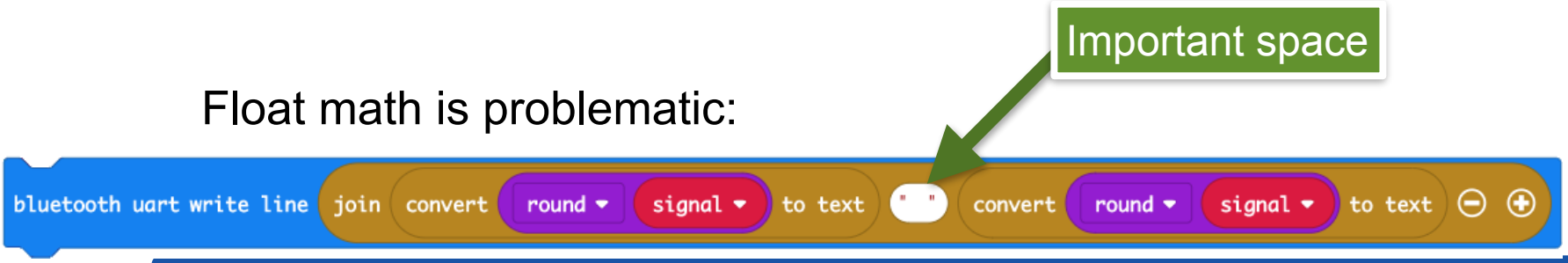




Instead use:




(found under Led)

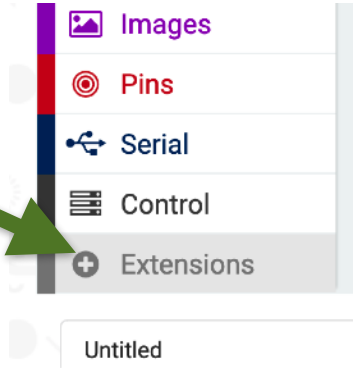
Float math is problematic:

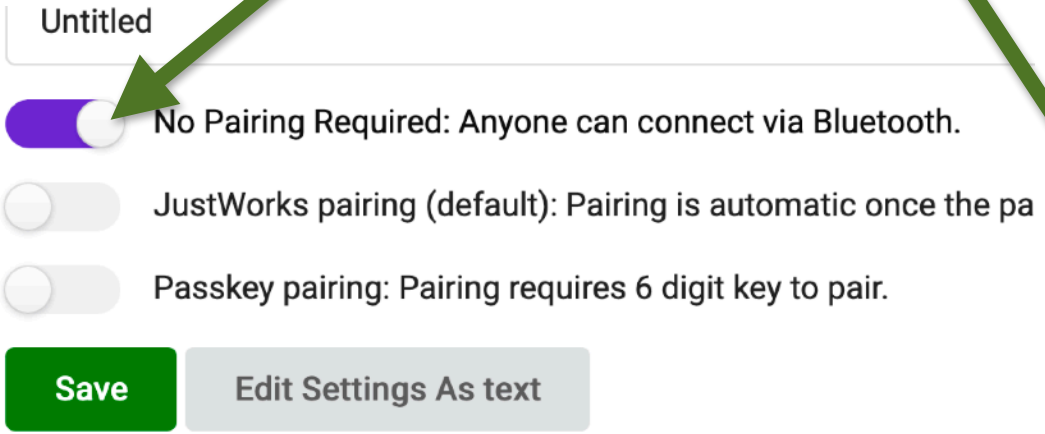


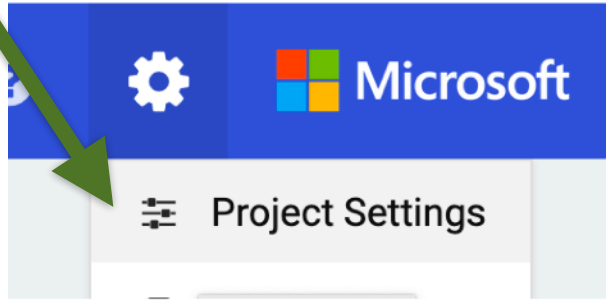
# How to make Bluetooth work

Select Extensions and add Bluetooth support

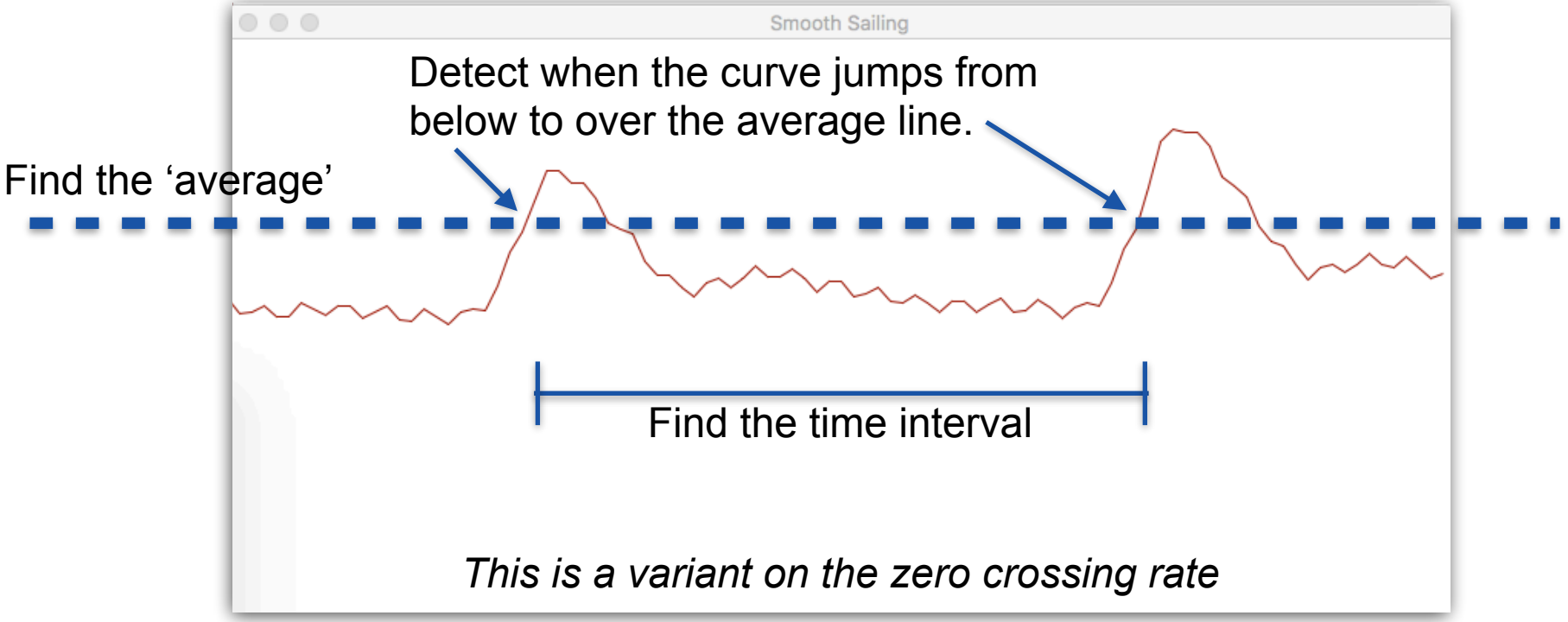
Select Project Settings and choose “No Pairing”



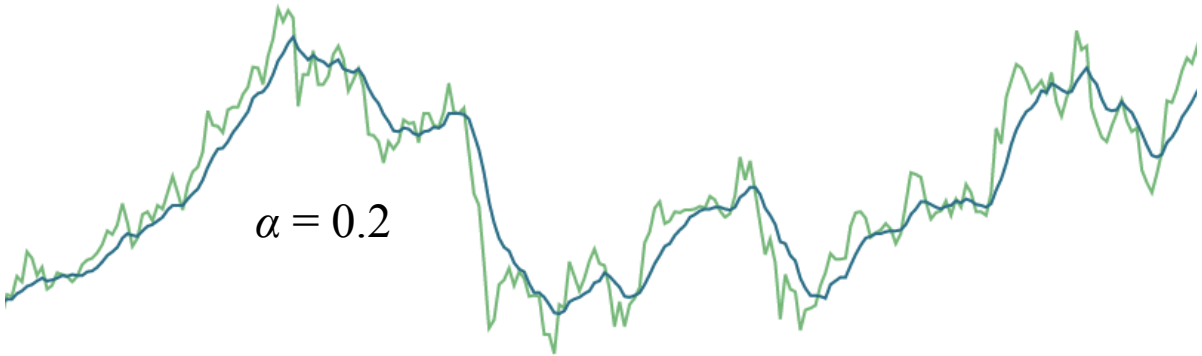
# One Possible Principle for a Solution



# EWMA

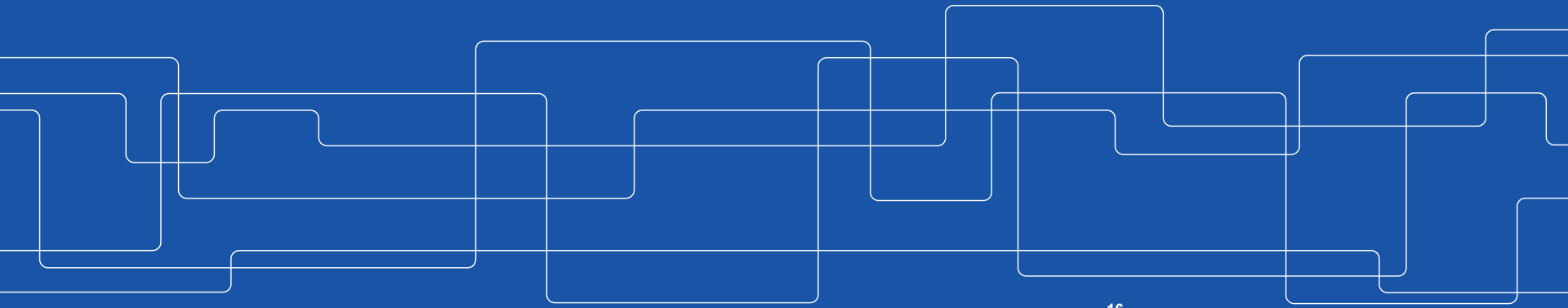
Exponentially Weighted Moving Average

$$S_t = \alpha \cdot Y_t + (1 - \alpha) \cdot S_{t-1} \qquad S_1 = Y_1$$



# Extra Stuff

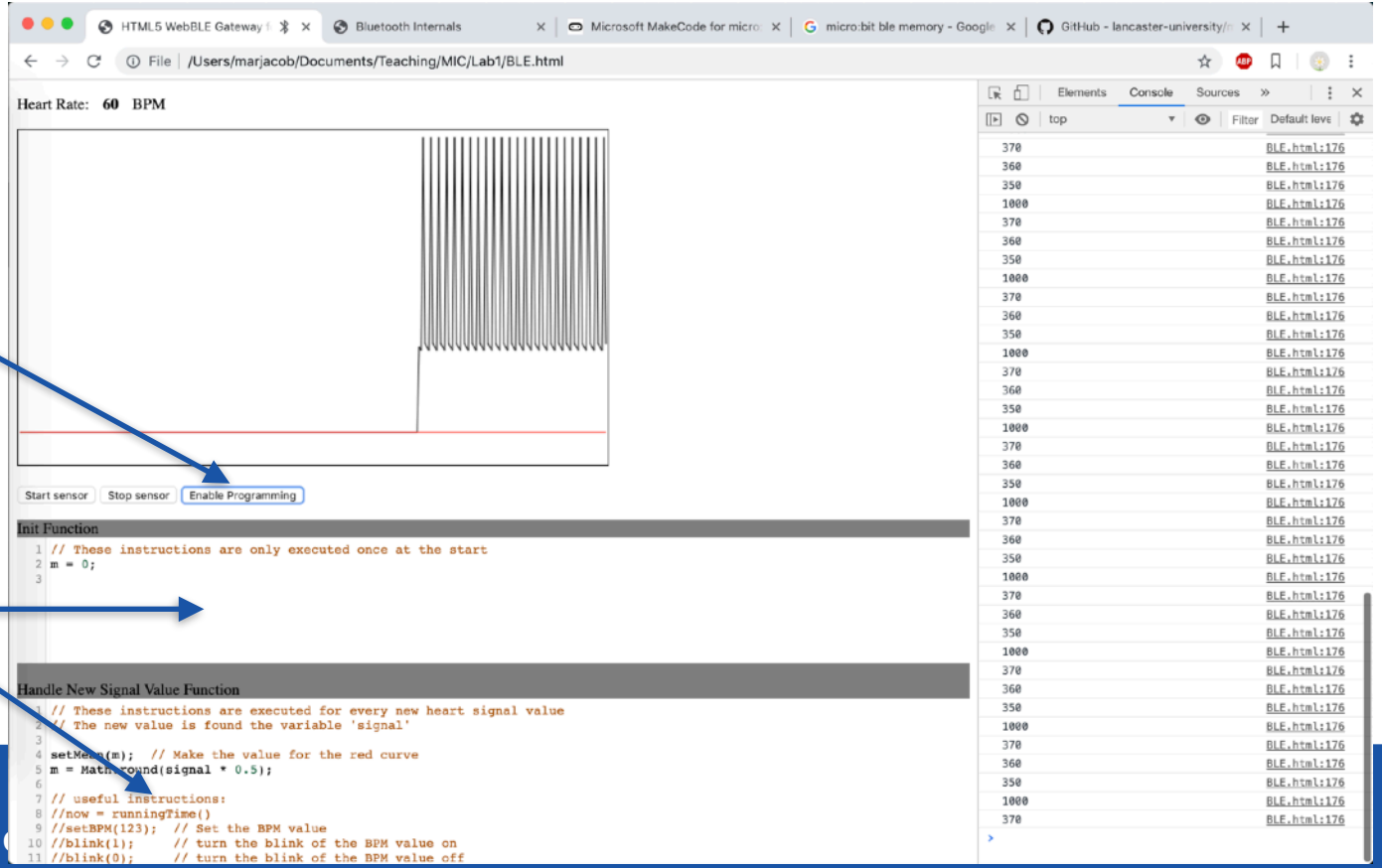
Not really needed …



# Move the Code to the Web!

Click this button

Paste and change the code here



# Use the MAX30102

Connect to device to the I2C on the Micro:Bit according to this link:

https://learn.sparkfun.com/tutorials/sparkfun-gatorparticle-hookup-guide/all



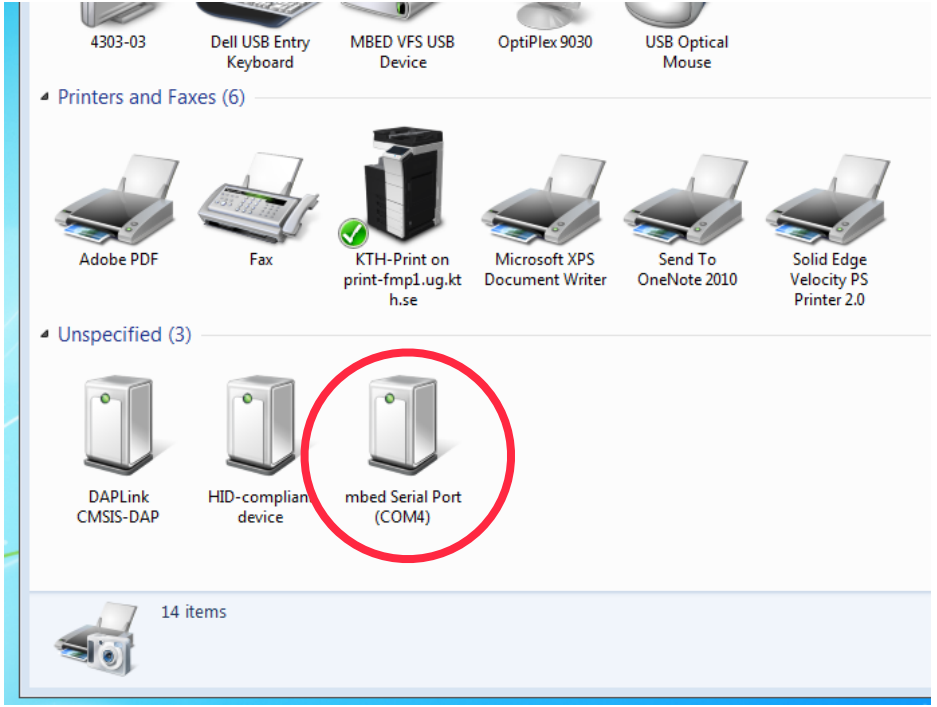

Then use this extension in MakeCode:

https://github.com/tinnal/pxt-gator-particle/

(The original gator particle extension does not work on newer Micro:Bit)

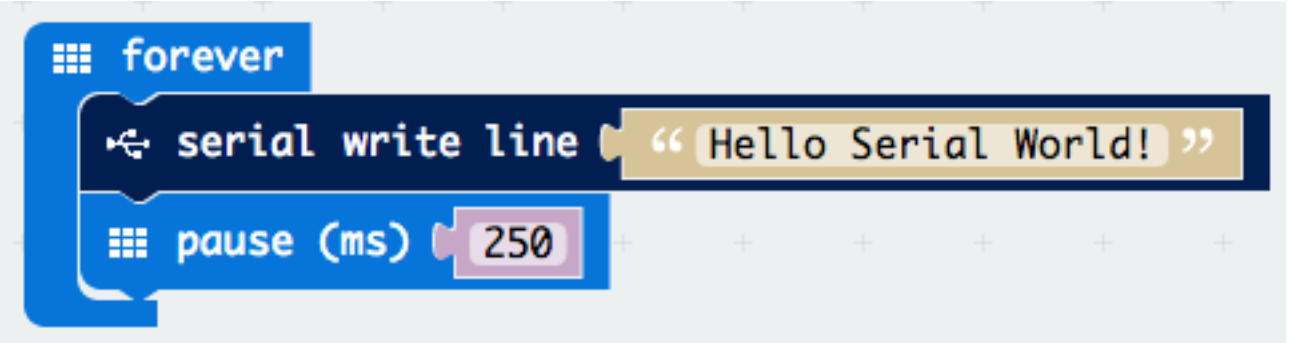
# Try the Serial Connection

- Find the COMx port
- Typically COM4 on our lab computers
- Choose the  button
- Devices and Printers



# Try the Serial Connection (cont:d)

- Create a new project like this:




- Then program the Micro:bit as before.
- On the computer: start **`putty`** from the Start menu (use the search field)

# Try the Serial Connection (cont:d)

- Connection type: ‘Serial’
- Serial Line: the COMx port
- Speed:115200
- Click ‘Open’

- You should see the text being printed on the screen over and over and over and over again
- Stop putty

